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ECEN 404
Status Update 3

Group #17: Patent Miner

List of team members:
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CJ Vines
Kosi Ezewudo
TA: Sabyasachi Gupta
Sponsor: Dr. Reddy Narasimha

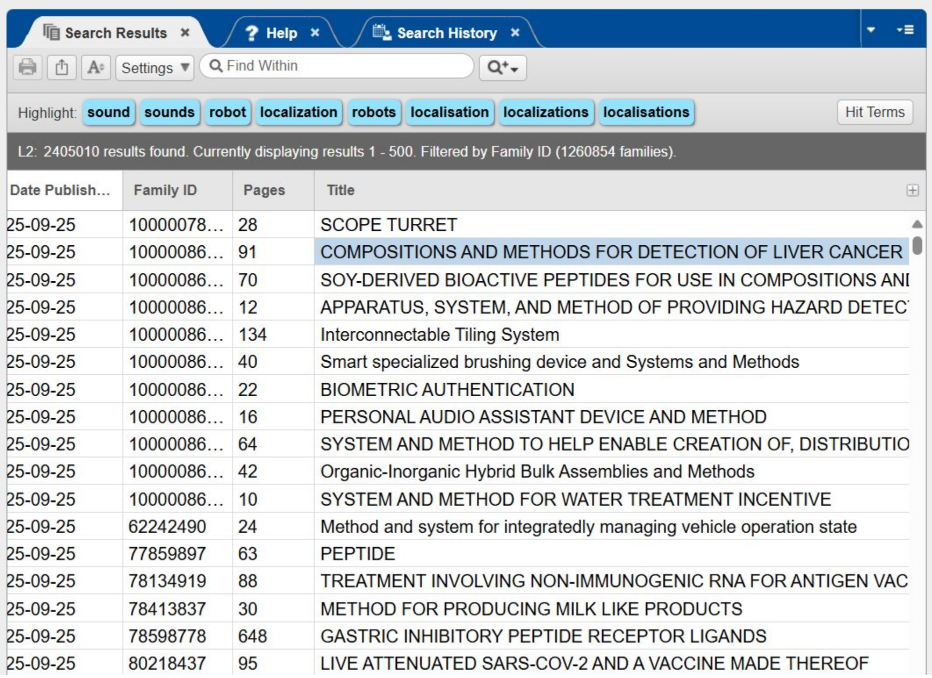
Project Summary

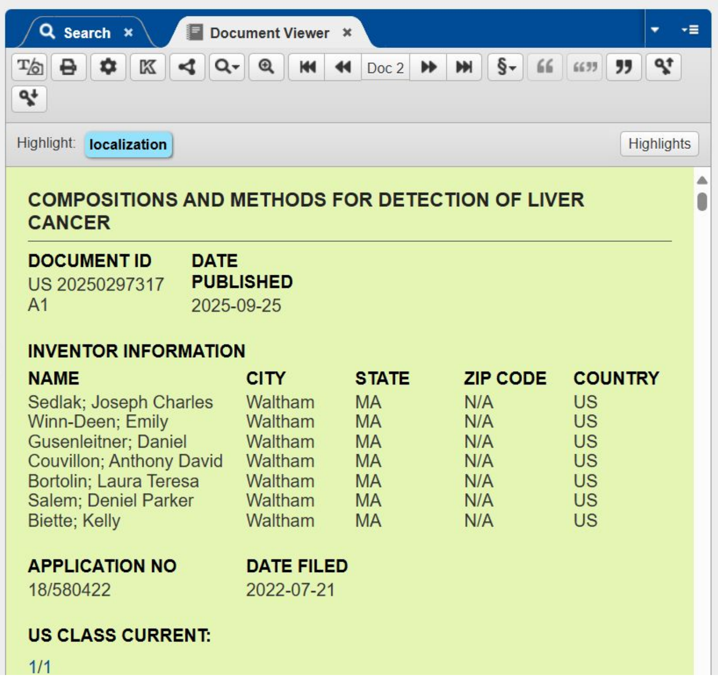
Problem statement:

- The current keyword search implemented by the United States Patent and Trademark Office (USPTO) fails to capture contextual meaning, leading to inaccuracy of search results and expensive billing rates for litigation professionals.

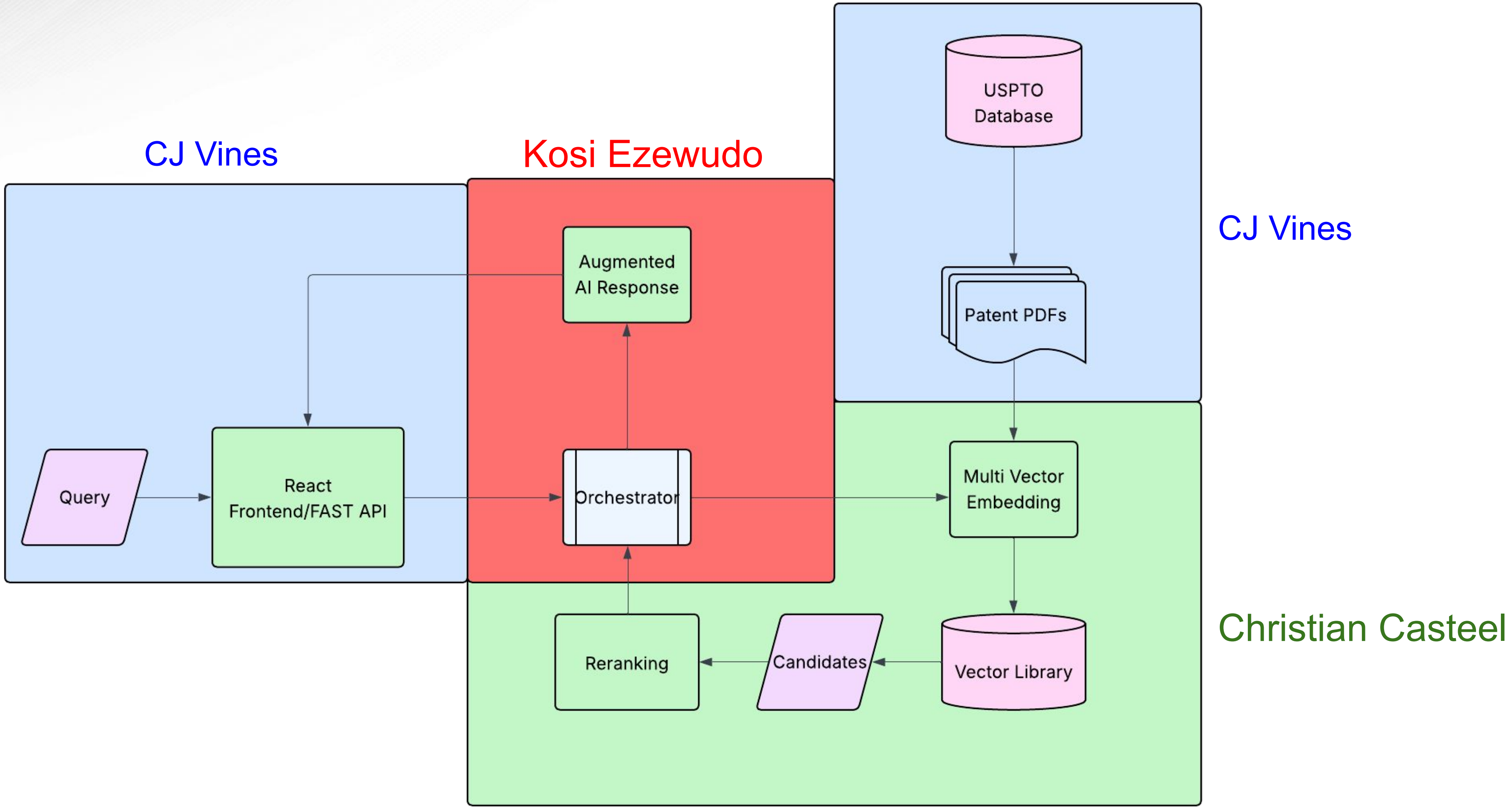
Solution:

- Development of a patent database mining tool that optimizes the patent search process by retrieving contextually relevant patents faster and more accurately compared to a traditional database.





System Overview



Project Timeline

1/12 — **~~Project Scope~~**

~~Revisit Subsystems, Define changes, Set dates~~

1/26 — **~~Subsystem Updates Completed~~**

~~Re-validate using 403 Validation Plan~~

2/9 — **~~Initial Integration + Debugging~~**

~~Identify potential issues, update ICD on any changes~~

2/16 — **Final System First Draft**

End-to-end integration completed, begin retrieval tweaking

3/2 — **Final System Validation**

Ensure all functional requirements are met

3/30 — **Completed System/Additional Features**

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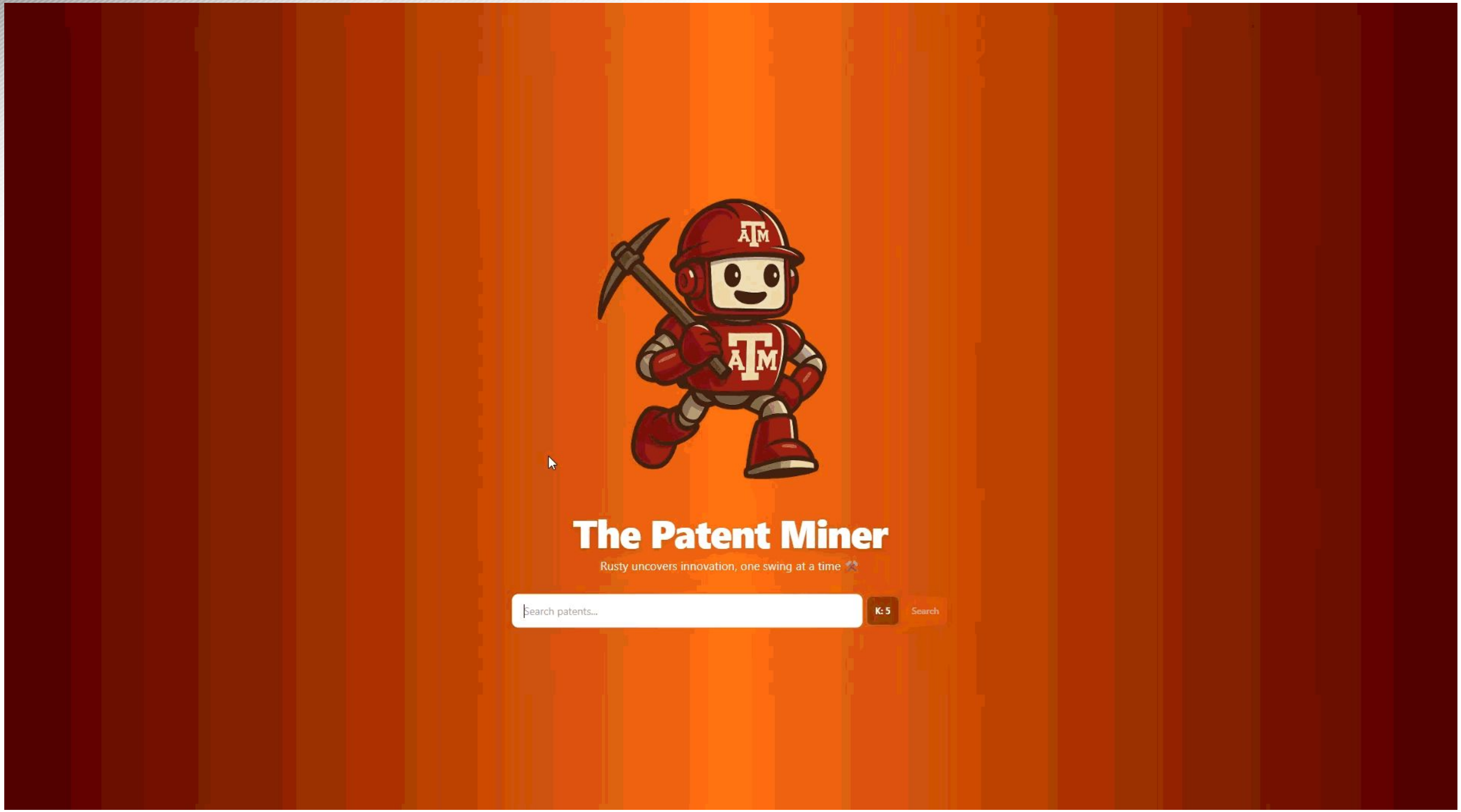


Embedding Pipeline and Storage

Christian Casteel

Accomplishments since last update	Ongoing progress/problems and plans until the next presentation
• Performed validation and discovered flaw in MUVERA implementation • Found queries and validation techniques to be unsatisfactory	• Form new validation query set with information from survey • Create more robust validation system, and rerun validation

Embedding Pipeline and Storage

Christian Casteel

	Metrics (768_f16 Shard)			
Method	P@10	R@10	nDCG@10	MRR@10
6k Patents	0.11	0.413	0.413	0.479
50k Patents	0.038	0.375	0.3	0.274
Accuracy Drop	65.50%	9.20%	43.00%	27.00%

Accuracy Degradation with Expanded Dataset

Score Relevancy automatically based on CPC:

Given G06F 7/00:

- **G06F 7/00** - perfect match - score of 3
- **G06F 7**/04 - main group match - score of 2
- **G06F** 13/00 - classification match - score of 1
- score of 0 otherwise

Shard	Size (6k Patents)	Size (50k Patents)	Size (Relative)
768_f32	23.2GB	193.3GB	100%
768_f16	11.7GB	97.5GB	50.40%
768_i8	5.9GB	48.7GB	25.20%
128_f32	3.3GB	27.5GB	14.20%
128_f16	1.8GB	15GB	7.80%

Approximate Database Sizes for Each Shard

Frontend/Preprocessing

CJ Vines

Accomplishments since last update	Ongoing progress/problems and plans until the next presentation
• Embedded 50k patents • Small front end update • CPC decoder • Debugged retrieval	• Contacting patent filers/litigation officials • Dynamic front end • Possible DB rework (again) • Further testing with 128 dim vectors

Weaviate Dilemma

- Weaviate uses a **H**ierarchical **N**avigable **S**mall **W**orlds (HNSW) graph to perform **A**pproximate **N**earest **N**eighbor (ANN) search on MuVERA vectors
  - This index in memory is ~14 GB
    - I am at my upper limit for RAM
- Solutions
  - Shard vector DB into multiple containers
    - Slight increase in complexity at retrieval time
  - Experiment with 128 dim
    - approximately ~6 times more patents

Weaviate

Retrieval Augmented Generation (RAG) Subsystem

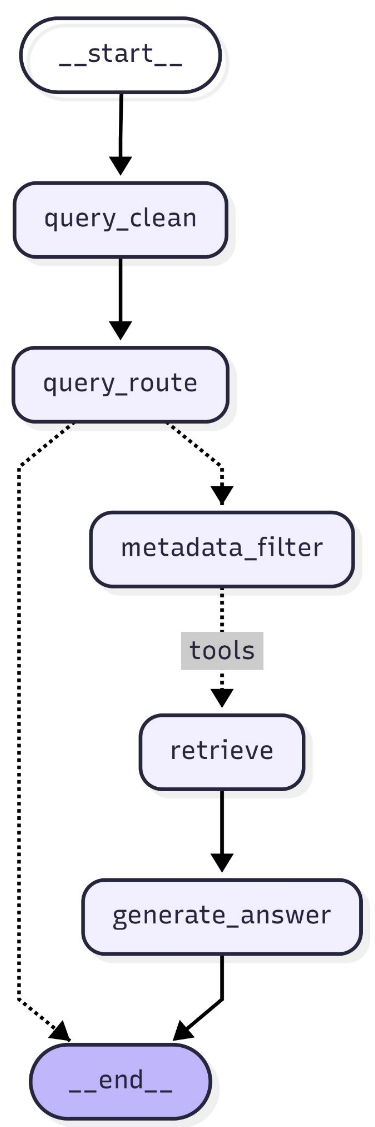
Kosi Ezewudo

Accomplishments since last update	Ongoing progress/problems and plans until the next presentation

• Performed Full-system Integration
• Began refining subsystem
   ○ Automated Validation Tests
   ○ Implemented Tracing for system end-end test visualization
   ○ Improving Query Metadata extraction
	• Continue Full system and Sub-system refinement
	   ○ Reference retrieved context in response
	• Brainstorm potential pathways for;
	   ○ Implementation of claims defense/combat dedicated layer.
	   ○ Visualization of retrieved patents and prior art
	• Perform extensive validation

Retrieval Augmented Generation Subsystem

Kosi Ezewudo



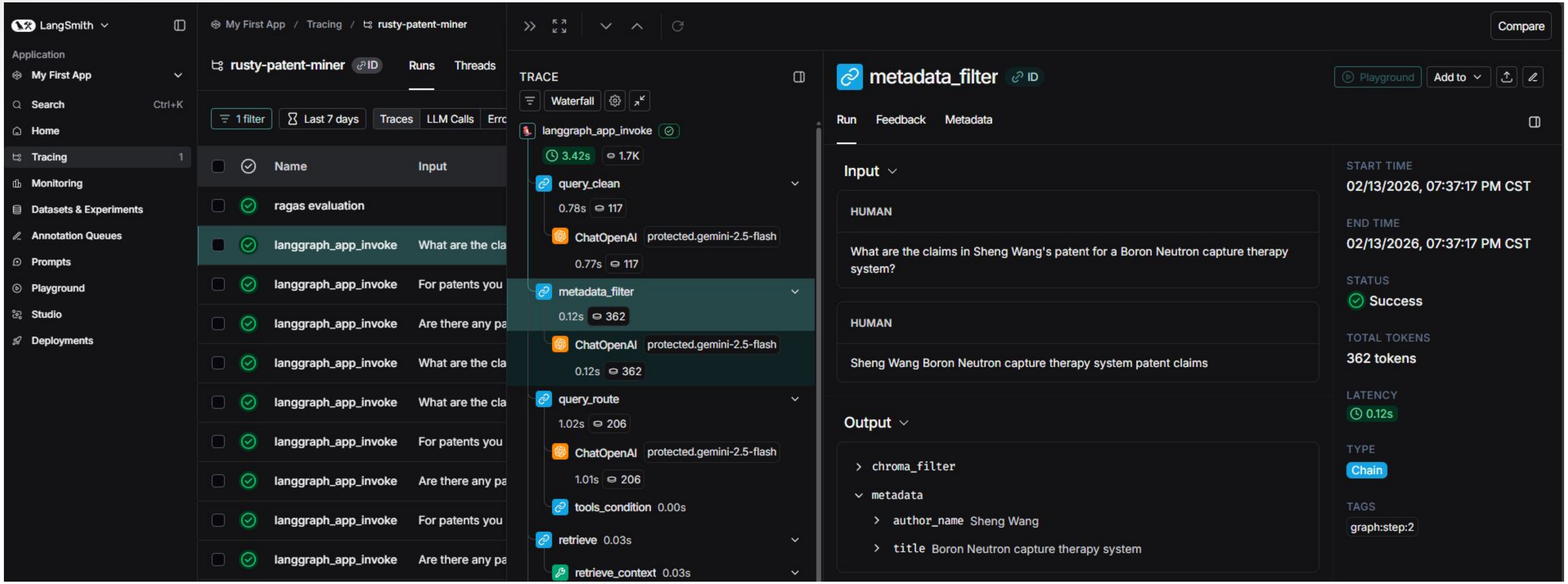


Fig 2. LangSmith Project Setup

Fig 1. Updated Backend Pipeline

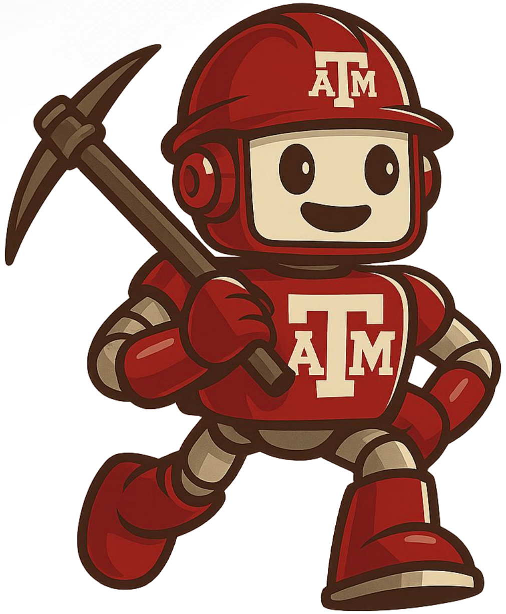
Execution Plan

	1/12/2026	1/19/2026	1/26/2026	2/2/2026	2/9/2026	2/16/2026	2/23/2026	3/2/2026	3/16/2026	3/23/2026	3/30/2026	4/6/2026	4/13/2026	4/20/2026	**Date**
Verify Multi Vector Storage as per 3.2.1.6.															1/19/2026
Finalize Metadata Extraction															1/26/2026
Implement Orchestrator															1/26/2026
Imeplement MUVERA															2/2/2006
Implement ColBERT reranking															2/2/2026
Test Semantic Chunking															2/2/2026
Validate Orchestrator as per 3.2.1.4.															2/9/2026
Validate Improved Retrieval as per 3.2.1.5.															2/9/2026
Backend - Orchestrator Integration															2/16/2026
Frontend - Backend Integration															2/23/2026
Full Integration															3/2/2026
Full Validation															4/6/2026
Trial Demo															4/20/2026
Final Demo															4/27/2026

Not Started
Behind Schedule
In Progress
Complete

Validation Plan

Paragraph #	Test Name	Success Criteria	Methodology	Status	Responsible Engineer(s)
3.2.1.2.	API/Pydantic Test	API gives correct response depending on the route taken 100% of the time	Send input query and data of different variations including edge cases like empty queries, queries with special characters, and get requests of the wrong shape	Tested	CJ Vines
3.2.1.1.	Determinism Test	Preprocessing Subsystem resulting chunks consistent across runs on the same dataset	Calculate SHA-256 hash of entire list of dictionaries after chunking and compare hash across runs to ensure download pipeline is deterministic. If any run is inconsistent, the test fails	Tested	CJ Vines
3.2.1.1.	Metadata Extraction	Stored metadata is verified to be at least 99% accurate	Compare metadata after chunking to metadata called from USPTO API, aiming for 99% accuracy	Tested	CJ Vines
3.2.1.4.	Full System Stress Test	Multiple simultaneous API calls are made successfully no data loss within 1ms	Make multiple API calls simultaneously, handle each concurrently in order.	Untested	CJ Vines & Kosi Ezewudo & Christian Casteel
3.2.1.5.	Embedding Reranking Test	"Golden" query - chunk pairs must have ≥ .05 Recall@k, Mean Reciprocal Rank (MRR) ≥ .5, and Normalized Discounted Cumulative Gain (NDCG) of .5	k = 10 target chunks at minimum with an equal number of matching queries. Calculate Recall@k, MRR, and NDCG for dataset. "Golden" set will include 50 hand selected patents.	Untested	Christian Casteel
3.2.1.5.	Embedding Accuracy Test	For a set of chunks, Hit@k of 1 and precision at k above .5. Relevance determined by patent id	Query containing information from multiple patents with similar CPC codes will test retrieval. Calculate Hit@k and Precision@k on top k chunks where k = 10.	Untested	Christian Casteel
3.2.1.1.	Failure Compliance Test	Pipeline can take in incorrect patent data and continue embedding without failure. Also will flag patents with false data	5 fake patents containing empty sections and incorrect CPC codes will be sent through the pipeline.	Untested	Christian Casteel
3.2.1.6.	Metadata Integrity Test	All stored metadata must be 100% consistent with metadata listed on a patent	Select 50 patents, specifically patents with generally unique CPCs and numbers of authors, then compare all metadata fields to the same patents on the USPTO database	Untested	Christian Casteel
3.2.1.1.	Download and Embedding Pipeline Stress Test	A USPTO bulk dataset is downloaded, embedded, and stored without failure	A USPTO bulk dataset download is conducted via fronetend download script. The bulk dataset should contain minimum 1000 patents.	Untested	Christian Casteel & CJ Vines
3.2.1.6.	Query Metadata Extraction	Extract all relevant metadata from user query, and use the metadata to filter patent search	Execute patent search using queries containing various metadata including prior date, patent ID, CPC etc.	Untested	Kosi Ezewudo
3.2.1.3.	Query Cleaning	Removes typos, sentence fragments etc. from user query	Execute patent search using grammatically incorrect queries	Untested	Kosi Ezewudo
3.2.1.4.	Generation Accuracy and Quality Test	Generated responses must have ≥ .8 Answer relevancy scores, ≥ .8 Diversity and Novelty	Query Patent Miner using golden set of patents. Generate responses and calculate Answer relevancy, Diversity and Novelty scores	Untested	Kosi Ezewudo & Christian Casteel & CJ Vines



Thank you!